nicknamed Alice and Bob – had to exchange encrypted messages, while a third computer named Eve tried to break their encryption. If Eve broke the encryption within a given time period, it got points. If it failed, Alice and Bob scored. After about fifteen thousand exchanges, Alice and Bob came up with a secret code that Eve couldn't break. Crucially, the Google engineers who conducted the experiment had not taught Alice and Bob anything about how to encrypt messages. The computers created a private language all on their own.[42]

Similar things are already happening in the world outside research laboratories. For example, the foreign exchange market (forex) is the global market for exchanging foreign currencies, and it determines the exchange rates between, say, the euro and the US dollar. In April 2022, the trade volume on the forex averaged $7.5 trillion per day. More than 90 per cent of this trading is already done by computers talking directly with other computers.[43] How many humans know how the forex market operates, let alone understand how the computers agree among themselves on trades worth trillions – and on the value of the euro and the dollar?

For the foreseeable future, the new computer-based network will still include billions of humans, but we might become a minority. For the network will also include billions – perhaps even hundreds of billions – of superintelligent alien agents. This network will be radically different from anything that existed previously in human history, or indeed in the history of life on earth. Ever since life first emerged on our planet about four billion years ago, all information networks were organic. Human networks like churches and empires were also organic. They had a lot in common with prior organic networks like wolf packs. They all kept revolving around the traditional biological dramas of predation, reproduction, sibling rivalry and romantic triangles. An information network dominated by inorganic computers would be different in ways that we can hardly even imagine. After all, as human beings, our imaginations are also products of organic biochemistry and cannot go beyond our preprogrammed biological dramas.

It has been only eighty years since the first digital computers were built. The pace of change is constantly accelerating, and we are nowhere close to exhausting the full potential of computers.[44] They may continue to evolve for millions of years, and what happened in the past eighty years is nothing compared with what's in store. As a crude analogy, imagine that we are in ancient Mesopotamia, eighty years after the first person thought of using a